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Assignment on

SIEM Sumulation: FIM, Brute Force Detection & Endpoint Vulnerability Assessment

Course: Enterprise Security Architecture Design and Management
Department of Computer Science and Engineering
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1. VM Configurations

Screenshot 1.1: VM Configuration — SIEM Server (Hostname, IP)

```
panda@SIEM-SERVER-02:~$ hostname
SIEM-SERVER-02
panda@SIEM-SERVER-02:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:12:cb:97 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.130/24 brd 192.168.2.255 scope global dynamic noprefixroute enp2s0
        valid_lft 1334sec preferred_lft 1334sec
    inet6 fe80::20c:29ff:fe12:cb97/64 scope link
        valid_lft forever preferred_lft forever
panda@SIEM-SERVER-02:~$
```

Screenshot 1.2: VM Configuration — Linux Agent (Hostname, IP)

```
panda@Agent-02:~$ hostname
Agent-02
panda@Agent-02:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:95:ec:7d brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.131/24 brd 192.168.2.255 scope global dynamic noprefixroute enp2s0
        valid_lft 1157sec preferred_lft 1157sec
    inet6 fe80::20c:29ff:fe95:ec7d/64 scope link
        valid_lft forever preferred_lft forever
panda@Agent-02:~$
```

Screenshot 1.3: Ping/Connectivity Test Between SIEM Server and Agents

```
panda@SIEM-SERVER-02:~$ ping 192.168.2.131
PING 192.168.2.131 (192.168.2.131) 56(84) bytes of data.
64 bytes from 192.168.2.131: icmp_seq=1 ttl=64 time=0.588 ms
64 bytes from 192.168.2.131: icmp_seq=2 ttl=64 time=0.201 ms
64 bytes from 192.168.2.131: icmp_seq=3 ttl=64 time=0.510 ms
64 bytes from 192.168.2.131: icmp_seq=4 ttl=64 time=0.591 ms
64 bytes from 192.168.2.131: icmp_seq=5 ttl=64 time=0.498 ms
64 bytes from 192.168.2.131: icmp_seq=6 ttl=64 time=0.510 ms
64 bytes from 192.168.2.131: icmp_seq=7 ttl=64 time=0.607 ms
64 bytes from 192.168.2.131: icmp_seq=8 ttl=64 time=0.526 ms
64 bytes from 192.168.2.131: icmp_seq=9 ttl=64 time=0.539 ms
64 bytes from 192.168.2.131: icmp_seq=10 ttl=64 time=0.370 ms
^C
--- 192.168.2.131 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9247ms
rtt min/avg/max/mdev = 0.201/0.494/0.607/0.116 ms
panda@SIEM-SERVER-02:~$
```

```
panda@Agent-02:~$
panda@Agent-02:~$ ping 192.168.2.130
PING 192.168.2.130 (192.168.2.130) 56(84) bytes of data.
64 bytes from 192.168.2.130: icmp_seq=1 ttl=64 time=0.390 ms
64 bytes from 192.168.2.130: icmp_seq=2 ttl=64 time=0.292 ms
64 bytes from 192.168.2.130: icmp_seq=3 ttl=64 time=0.203 ms
64 bytes from 192.168.2.130: icmp_seq=4 ttl=64 time=0.559 ms
64 bytes from 192.168.2.130: icmp_seq=5 ttl=64 time=0.492 ms
64 bytes from 192.168.2.130: icmp_seq=6 ttl=64 time=0.476 ms
64 bytes from 192.168.2.130: icmp_seq=7 ttl=64 time=0.525 ms
64 bytes from 192.168.2.130: icmp_seq=8 ttl=64 time=0.498 ms
^C
--- 192.168.2.130 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7131ms
rtt min/avg/max/mdev = 0.203/0.429/0.559/0.116 ms
panda@Agent-02:~$ █
```

1.2 System Resource Allocation

The SIEM server requires adequate computational resources to handle log ingestion, rule processing, and dashboard visualization. The following resource allocation ensures optimal performance:

Hostname	vCPU	RAM	Storage	Network Mode
SIEM-SERVER-02	4 vCPU	8 GB RAM	80 GB SSD	NAT/Host-Only
Agent-02	2 vCPU	4 GB RAM	20 GB SSD	NAT/Host-Only
Attacker-02	2 vCPU	4 GB RAM	60 GB SSD	NAT/Host-Only

2. SIEM Server Installation & Configuration

This section documents the complete installation and configuration of the Wazuh SIEM server (SIEM-SERVER). Wazuh is an open-source security platform that provides unified XDR (Extended Detection and Response) and SIEM (Security Information and Event Management) capabilities for threat detection, compliance monitoring, and incident response.

2.1 Wazuh Server Installation Steps

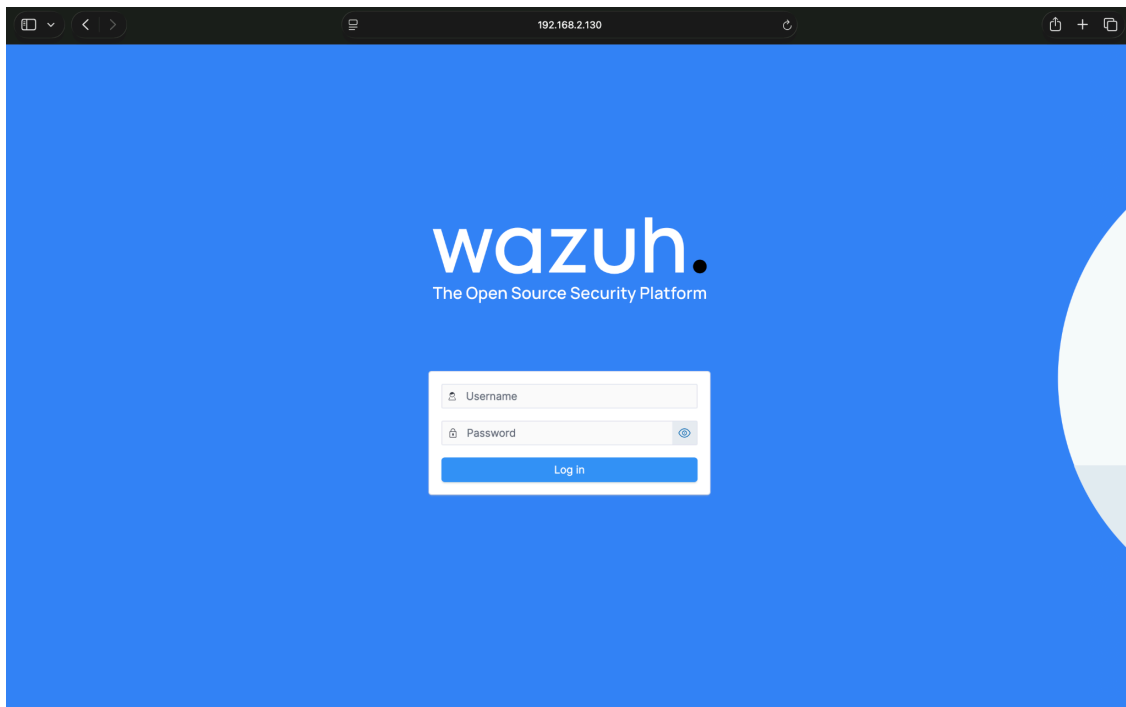
1. Update the system package repository: `sudo apt update && sudo apt upgrade -y`
2. Download and execute the Wazuh installation assistant script from the official repository.
3. Run the unattended installation: `sudo bash wazuh-install.sh -a` (all-in-one deployment including indexer, server, and dashboard).
4. The installation process automatically deploys: Wazuh Indexer (OpenSearch), Wazuh Server (analysis engine), and Wazuh Dashboard (Kibana-based visualization).
5. Upon completion, the script outputs the admin credentials and dashboard access URL (<https://192.168.2.130>).

6. Verify all Wazuh services are active: `sudo systemctl status wazuh-indexer, wazuh-manager, wazuh-dashboard`.

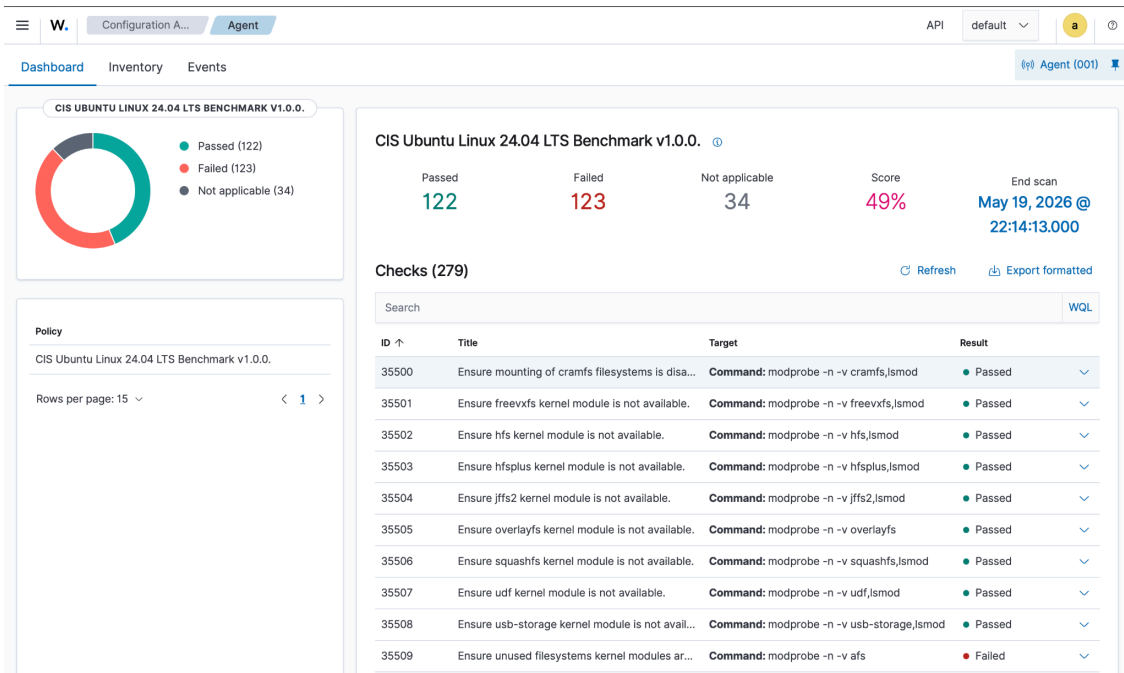
2.2 Wazuh Dashboard Access & Initial Configuration

After successful installation, access the Wazuh Dashboard via a web browser using the HTTPS protocol. The default port is 443. Log in using the auto-generated admin username and password. Complete the initial setup wizard to configure the deployment environment.

Screenshot 2.1: Wazuh Dashboard Login Page



Screenshot 2.2: Wazuh Dashboard Main Overview Screen



2.3 Agent Enrollment Configuration

Wazuh agents must be enrolled with the SIEM server to establish secure communication. The enrollment process uses the Wazuh agent enrollment protocol over port 1515 (authd service). Configure the server to accept agent connections and generate deployment commands for each endpoint.

Server-side enrollment settings are configured in `/var/ossec/etc/ossec.conf`. Ensure the `authd` section allows agent enrollment with the correct password or SSL certificate authentication.

Screenshot 2.3: Wazuh Agent Enrollment Settings (ossec.conf)

```
<!--
Wazuh - Agent - Default configuration for ubuntu 24.04
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <client>
    <server>
      <address>192.168.2.130</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>ubuntu, ubuntu24, ubuntu24.04</config-profile>
    <notify_time>20</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
    <crypto_method>aes</crypto_method>
  </client>

  <client_buffer>
    <!-- Agent buffer options -->
    <disabled>no</disabled>
    <queue_size>5000</queue_size>
    <events_per_second>500</events_per_second>
  </client_buffer>
```

2.4 Firewall & Network Security Configuration

The following ports must be open on the SIEM server to ensure proper communication between components:

Port/Protocol	Purpose
1514/TCP	Wazuh Agent → Server (secure event forwarding)
1515/TCP	Wazuh Agent Enrollment (authd service)
55000/TCP	Wazuh API (management and queries)
443/TCP	Wazuh Dashboard (HTTPS web interface)
9200/TCP	Wazuh Indexer (internal, localhost only)

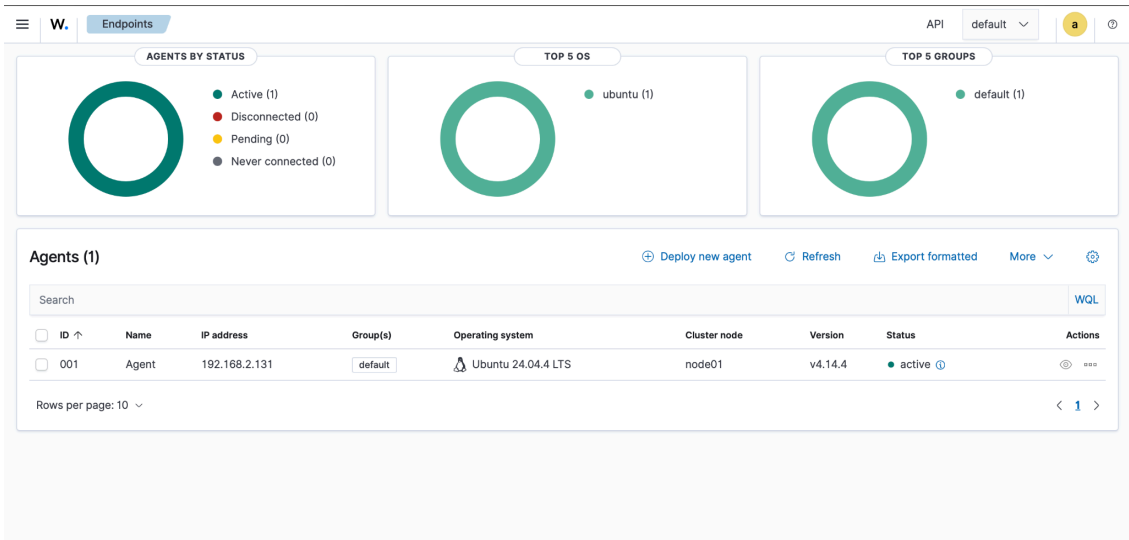
3. Q1: Endpoint Integration & Vulnerability Detection

This section demonstrates the integration of endpoint agents with the Wazuh SIEM server and the configuration assessment and vulnerability detection capabilities. Evidence of successful agent enrollment, configuration policy compliance, and vulnerability scan results is documented.

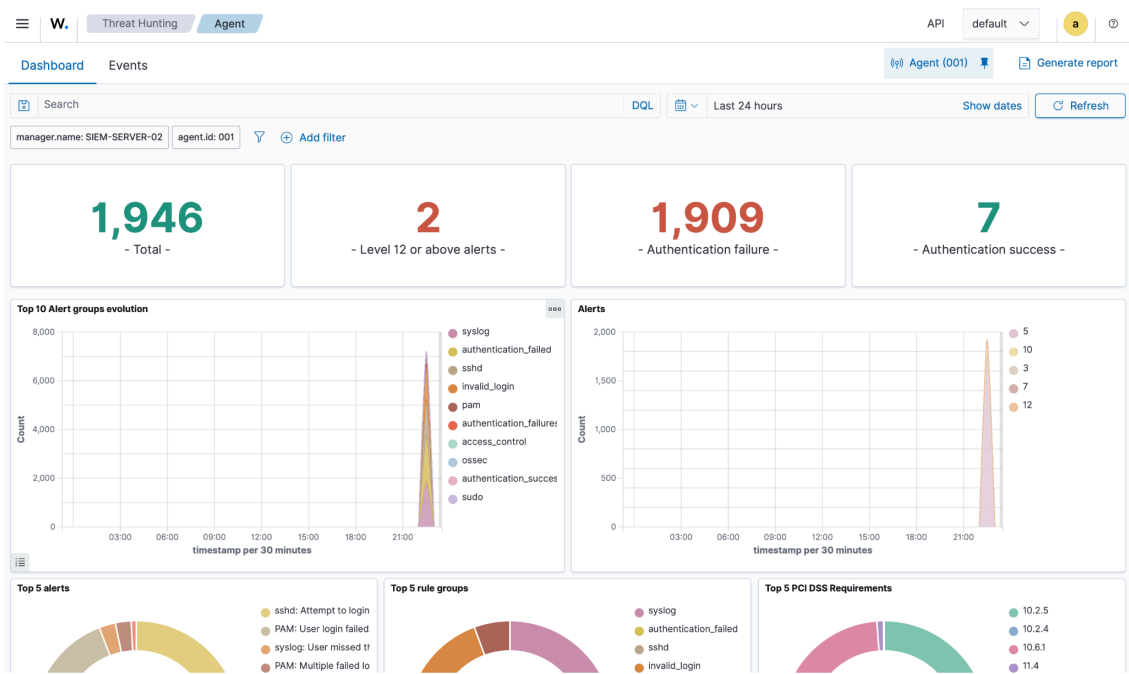
3.1 Evidence of Agent Integration — Dashboard View

The Wazuh Dashboard provides a centralized view of all enrolled agents, their operational status, OS platform, version, and last keep-alive timestamp. This serves as primary evidence of successful endpoint integration.

Screenshot 3.1: Wazuh Dashboard — Agents List Showing Both Endpoints Active



Screenshot 3.2: Agent Details Page — Agent-02 (Linux) Overview



4. Q2: File Integrity Monitoring (FIM) & Malware Detection

This section demonstrates the File Integrity Monitoring (FIM) capability of Wazuh to detect unauthorized file modifications, including the placement of a test virus/malware file. The FIM module uses Syscheck to monitor specified directories in real-time and generates alerts when files are created, modified, or deleted. Additionally, VirusTotal integration is configured as a threat intelligence feed to enrich alerts with malware detection data.

4.1 FIM Module Configuration on SIEM Server

The FIM module (Syscheck) is configured in the Wazuh manager's `ossec.conf` file. Directories to monitor are specified with `<directories>` tags, and real-time monitoring is enabled using the `realtime='yes'` attribute. The following configuration monitors the designated folder for this exercise:

```
<directories check_all="yes" realtime="yes">/home/user/Folder-02</directories>
```

The `check_all='yes'` attribute enables monitoring of all file properties including size, permissions, ownership, modification time, inode, and MD5/SHA1/SHA256 hash values. The `realtime='yes'` attribute enables inotify (Linux).

Screenshot 4.1: FIM Configuration in `/var/ossec/etc/ossec.conf`

```
<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <scan_on_start>yes</scan_on_start>

  <!-- Directories to check (perform all possible verifications) -->
  <!-- <directories>/etc,/usr/bin,/usr/sbin</directories> -->
  <!-- <directories>/bin,/sbin,/boot</directories> -->
  <directories check_all="yes" report_changes="yes" realtime="yes">/home/panda/Folder-02</directories>

  <!-- Files/directories to ignore -->
  <ignore>/etc/mtab</ignore>
  <ignore>/etc/hosts.deny</ignore>
  <ignore>/etc/mail/statistics</ignore>
  <ignore>/etc/random-seed</ignore>
```

Screenshot 4.2: FIM Directory Verification — Folder-02 Exists on Agent

```
panda@Agent-02:~$ ls
Desktop  Downloads  Folder-02  Pictures  Templates  snap          wazuh-agent_4.14.4-1_amd64.deb
Documents FIM        Music      Public    Videos    virus-total-file

panda@Agent-02:~$ cd Folder-02/
panda@Agent-02:~/Folder-02$ ls
note.txt testvirus.txt
panda@Agent-02:~/Folder-02$
```

4.2 VirusTotal Integration as Threat Intelligence Feed

VirusTotal is a widely used threat intelligence platform that aggregates antivirus scan results from multiple vendors. Wazuh integrates with the VirusTotal API v3 to automatically query file hashes and enrich FIM alerts with malware detection verdicts. This integration requires a valid VirusTotal API key.

1. Register for a free VirusTotal API key at <https://www.virustotal.com>.
2. Configure the VirusTotal integration block in `/var/ossec/etc/ossec.conf` on the SIEM server.
3. Set the API key: `<api_key>VIRUSTOTAL_API_KEY</api_key>`.
4. Configure the `rule_id` filter to only query FIM-related alerts (`rule.groups` contains `syscheck`).
5. Set the alert level threshold: `<group>syscheck</group>` with `<alert_format>json</alert_format>`.
6. Restart the Wazuh manager to activate the integration.

```
<integration>
  <name>virustotal</name>
  <api_key>YOUR_VIRUSTOTAL_API_KEY</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>
```

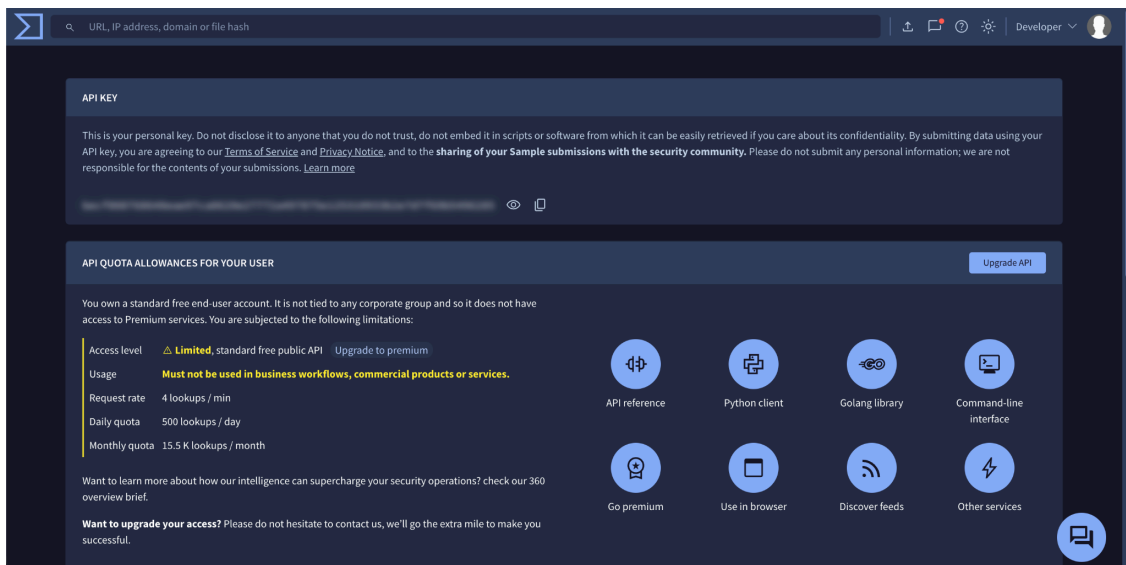
Screenshot 4.3: VirusTotal Integration Configuration in `ossec.conf`

```
<integration>
  <name>virustotal</name>
  <api_key>becf808768648eae97ca0620e27772a497875e125310933b2e7d7f69b5496285</api_key> <!-- Replace with your VirusTotal API key -->
  <!-- <rule_id>554,550</rule_id> -->
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>

</ossec_config>

<ossec_config>
  <localfile>
    <log_format>journaId</log_format>
    <location>journaId</location>
  </localfile>
```

Screenshot 4.4: VirusTotal API Key Registration Page



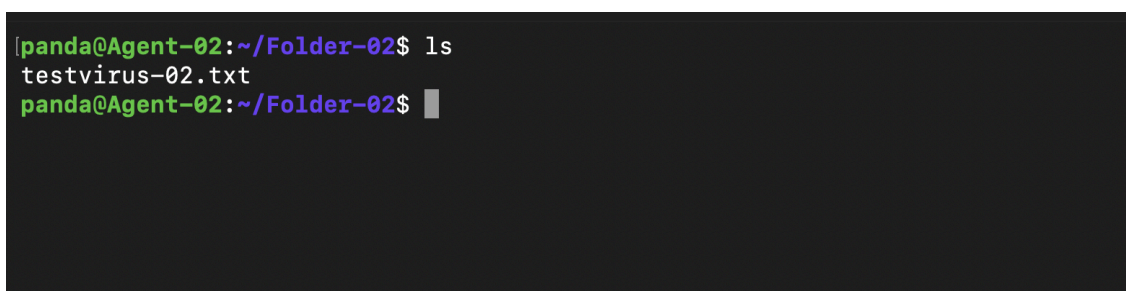
4.3 Test Virus/Malware File Preparation

To simulate a malware detection scenario, a test virus file is created in the monitored directory. For safe testing, the EICAR (European Institute for Computer Antivirus Research) standard anti-malware test file is used. This file contains a standardized string that all antivirus engines recognize as a test virus without causing actual harm.

The test file is named testvirus-02.txt as per the assignment requirements and placed in the Folder-02 directory.

EICAR Test String:
X5O!P%@AP[4\PZX54(P^)7CC)7{\$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!\$H+H*

Screenshot 4.5: Test Virus File Creation — testvirus-02.txt in Folder-02



Screenshot 4.6: File Content Verification (cat/nano showing EICAR string)

```
[panda@Agent-02:~/Folder-02$ ls
testvirus-02.txt
[panda@Agent-02:~/Folder-02$ cat testvirus-02.txt
X50!P%AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
[panda@Agent-02:~/Folder-02$
```

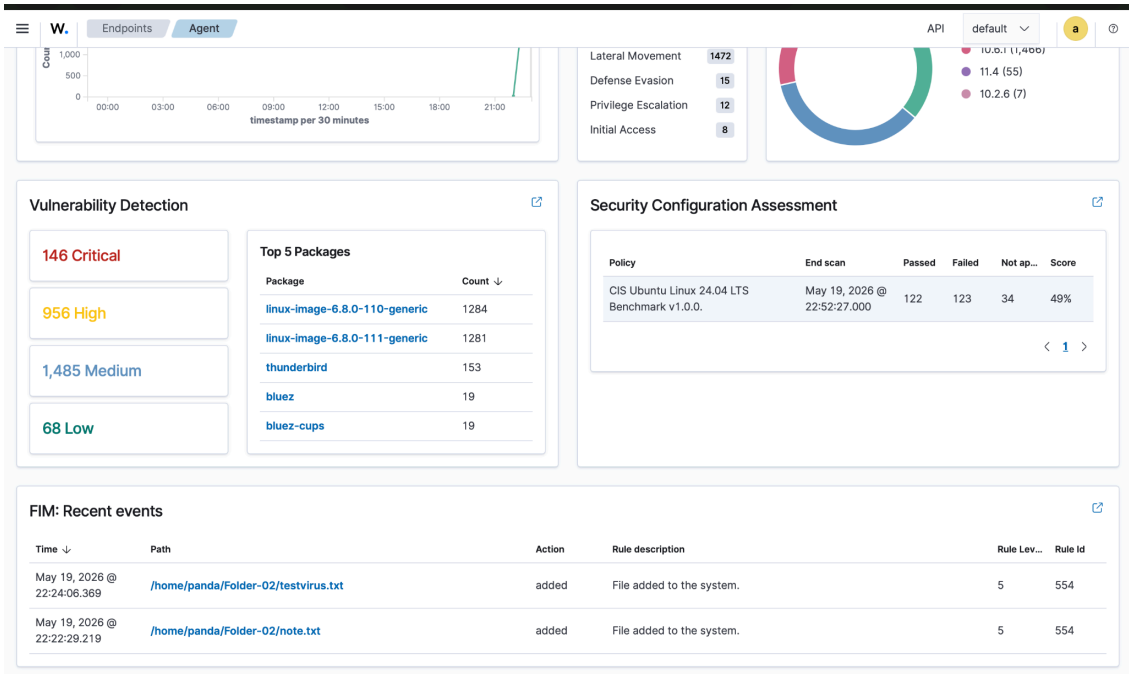
4.4 FIM Alert Generation & Dashboard Visualization

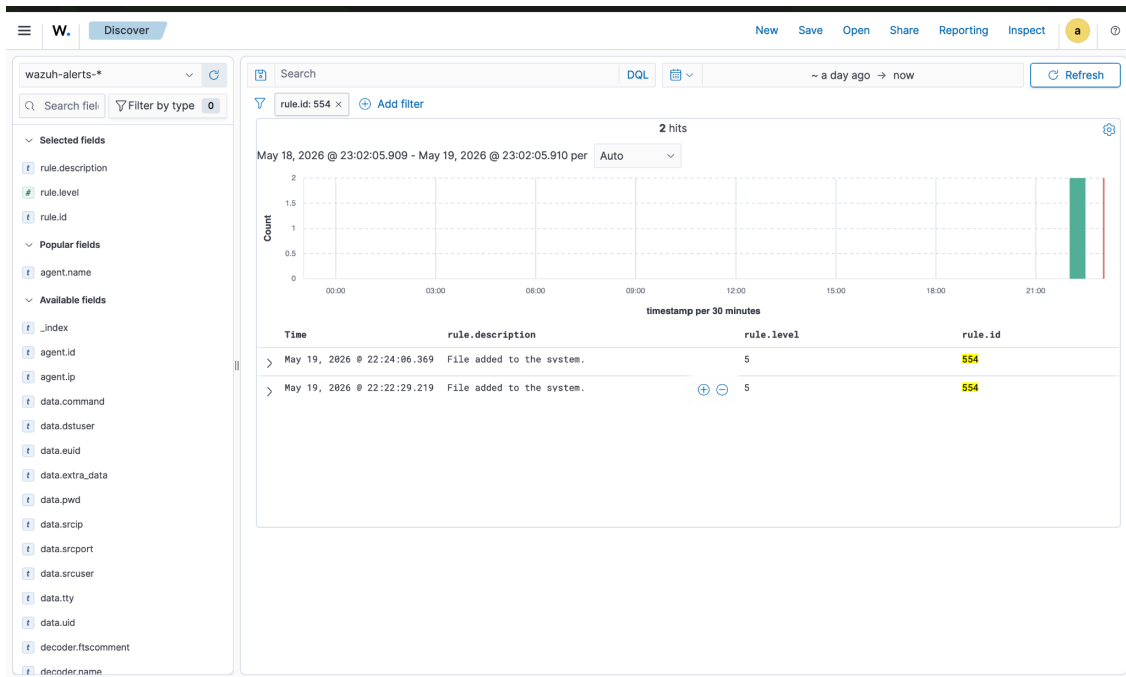
Upon creation of the testvirus-02.txt file, the FIM module detects the new file and generates a Syscheck alert. The alert includes file metadata (path, size, permissions, hash values) and triggers the VirusTotal integration to query the file hash against the VirusTotal database.

a) Relevant Alert on Dashboard

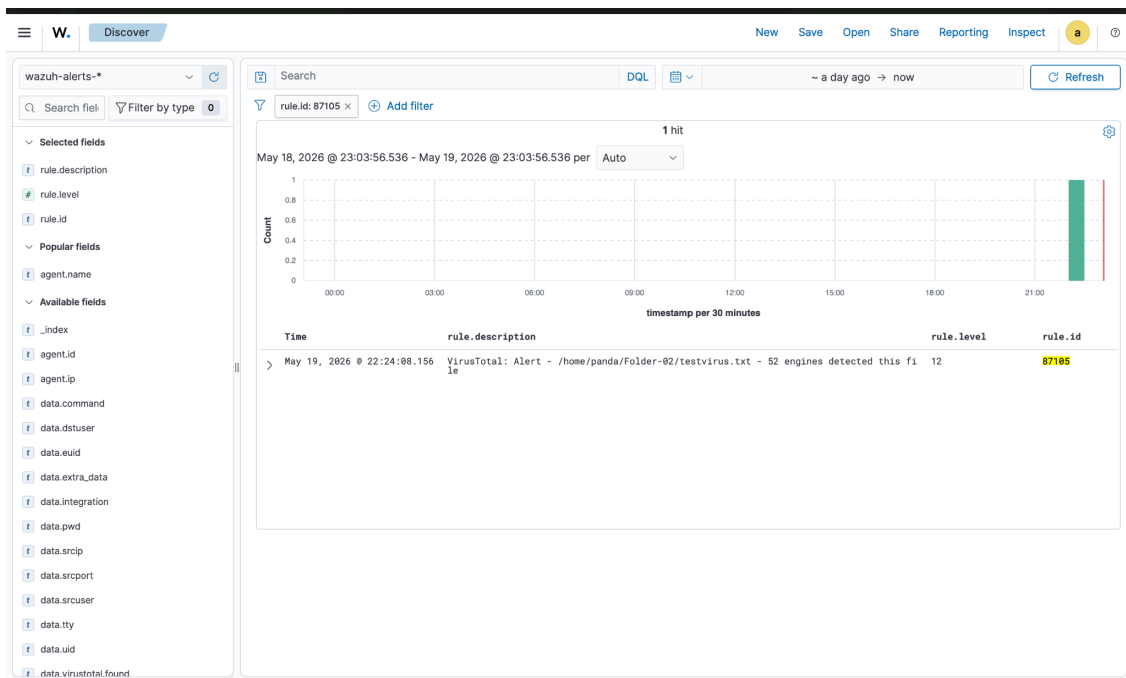
The Wazuh Dashboard displays the FIM alert under the Security Events module. The alert is categorized under the rule group 'syscheck' with a severity level reflecting the file addition event. The VirusTotal enrichment adds a secondary alert indicating the number of antivirus engines that detected the file as malicious.

Screenshot 4.7: Dashboard Alert — FIM File Added Event for testvirus.txt

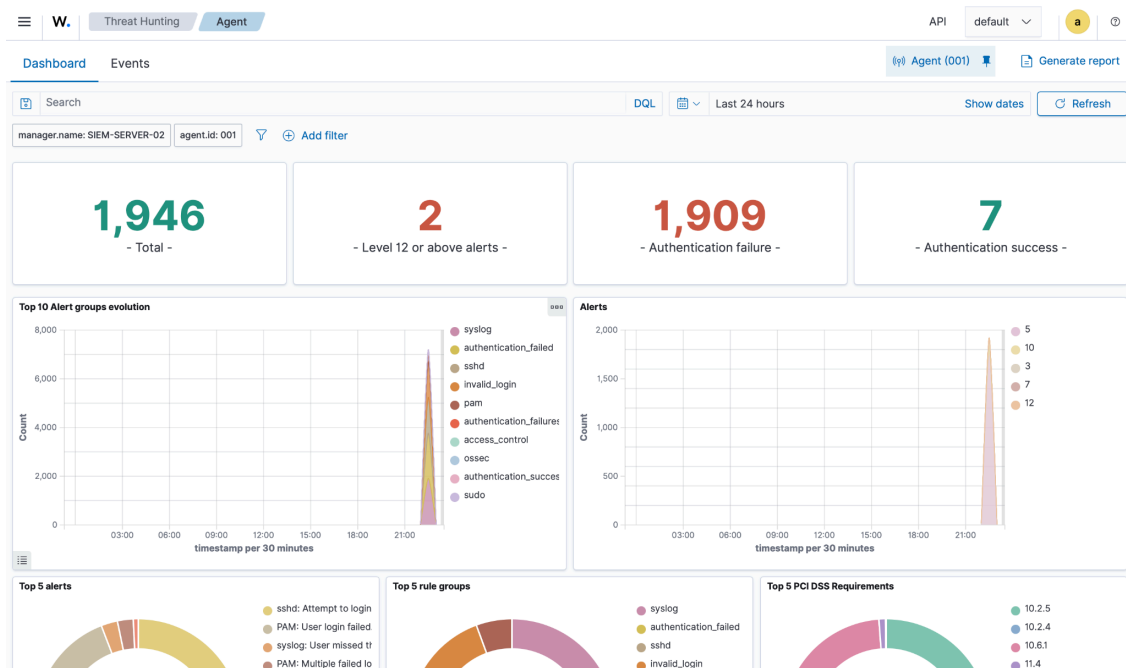




Screenshot 4.8: Dashboard Alert — VirusTotal Positive Detection Result



Screenshot 4.9: Dashboard Event Timeline — FIM and VirusTotal Alerts Correlated



b) Raw Alert Logs Snapshots from Server

Raw alert logs are stored in `/var/ossec/logs/alerts/alerts.json` on the SIEM server. These JSON-formatted logs contain the complete alert payload including timestamp, rule information, agent metadata, and file details.

Screenshot 4.10: Raw FIM Alert Log — alerts.json Entry for File Addition

```
root@SIEM-SERVER-02:/home/panda# cat /var/ossec/lib/ logs/
root@SIEM-SERVER-02:/home/panda# cat /var/ossec/logs/alerts/alerts.json | grep 'Folder-02'
{"timestamp":"2026-05-19T16:22:29.219+0000","rule":{"level":5,"description":"File added to the system.","id":"554","firedtimes":1,"mail":false,"groups":["ossec","syscheck","syscheck_entry_added","syscheck_file"],"pci_dss":["11.5"],"pgp13":["4.11"],"gdpr":["II_5.1.f"],"hipaa":["164.312.c.1","164.312.c.2"],"nist_800_53":["SI.7"],"tsc":["PI1.4","PI1.5","CC6.1","CC6.8","CC7.2","CC7.3"],"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":{"name":"SIEM-SERVER-02"},"id":"1779207749.20183","full_log":"File '/home/panda/Folder-02/note.txt' added\nMode: realtime\n","syscheck":{"path":"/home/panda/Folder-02/note.txt","mode":"realtime","size_after":"44","perm_after":"rw-rw-r--","uid_after":"1000","gid_after":"1000","md5_after":"d0796b7c8a6087aafa7decba683311ba","sha1_after":"1f44c8d9d6d879c551881f2bdc3c35c5adaf24ea","sha256_after":"6bc486e20fd0035c64afa3162bddf4b37742c6a4faca384cb589a5a33b0f7451","uname_after":"panda","gname_after":"panda","mtime_after":"2026-05-19T16:22:29","inode_after":"526380","event":"added"},"decoder":{"name":"syscheck_new_entry"},"location":"syscheck"}
```

Screenshot 4.11: Raw VirusTotal Integration Log — Threat Intelligence Enrichment

```
root@SIEM-SERVER-02:/home/panda# cat /var/ossec/logs/alerts/alerts.json | grep 'virustotal'
{"timestamp":"2026-05-19T16:22:31.818+0000","rule":{"level":3,"description":"VirusTotal: Alert - No records in VirusTotal database","id":"87103","firedtimes":1,"mail":false,"groups":["virustotal"],"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":{"name":"SIEM-SERVER-02"},"id":"1779207751.20882","decoder":{"name":"json"},"data":{"virustotal":{"found":"0","malicious":"0","source":{"alert_id":"1779207749.20183"},"file_path":"/home/panda/Folder-02/note.txt","md5":"d0796b7c8a6087aafa7decba683311ba","sha1":"1f44c8d9d6d879c551881f2bdc3c35c5adaf24ea"},"integration":"virustotal"},"location":"virustotal"}
{"timestamp":"2026-05-19T16:24:08.156+0000","rule":{"level":12,"description":"VirusTotal: Alert - /home/panda/Folder-02/testvirus.txt - 52 engines detected this file","id":"87105","mitre":{"id":"T1203"},"tactic":{"Execution"},"technique":{"Exploitation for Client Execution"},"firedtimes":1,"mail":true,"groups":["virustotal"],"pci_dss":["10.6.1","11.4"],"gdpr":["IV_35.7.d"],"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":{"name":"SIEM-SERVER-02"},"id":"1779207848.22308","decoder":{"name":"json"},"data":{"virustotal":{"found":"1","malicious":"1","source":{"alert_id":"1779207846.21604"},"file_path":"/home/panda/Folder-02/testvirus.txt","md5":"69630e4574ec6798239b091cda43dca0","sha1":"cf8bd9dfddff007f75adf4c2be48005cea317c62"},"scan_date":"2026-05-19 09:45:46","positives":"52","total":"58","permalink":"https://www.virustotal.com/gui/file/131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbfd8267/detection/f-131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbfd8267-1779183946"},"integration":"virustotal"},"location":"virustotal"}
root@SIEM-SERVER-02:/home/panda#
```

Screenshot 4.12: Wazuh Log Analysis — /var/ossec/logs/ossec.log Showing FIM Scan

```
-02.
2026/05/19 15:20:14 indexer-connector: INFO: IndexerConnector initialized successfully for index: wazuh-02.
2026/05/19 15:20:14 indexer-connector: INFO: IndexerConnector initialized successfully for index: wazuh-02.
2026/05/19 15:20:14 indexer-connector: INFO: IndexerConnector initialized successfully for index: wazuh-02.
2026/05/19 15:20:14 indexer-connector: INFO: IndexerConnector initialized successfully for index: wazuh-02.
2026/05/19 15:20:14 indexer-connector: INFO: IndexerConnector initialized successfully for index: wazuh-m-server-02.
2026/05/19 15:21:32 wazuh-logcollector: INFO: (9204): 'Journald' timestamp was refreshed due to rotation
2026/05/19 15:23:13 wazuh-modulesd:vulnerability-scanner: INFO: Initiating update feed process.
2026/05/19 15:23:56 wazuh-modulesd:vulnerability-scanner: INFO: Triggered a re-scan after content update
2026/05/19 15:23:56 wazuh-modulesd:vulnerability-scanner: INFO: Feed update process completed.
2026/05/19 16:19:14 wazuh-modulesd:syscollector: INFO: Starting evaluation.
2026/05/19 16:19:50 wazuh-modulesd:syscollector: INFO: Evaluation finished.
2026/05/19 17:19:51 wazuh-modulesd:syscollector: INFO: Starting evaluation.
2026/05/19 17:20:06 wazuh-modulesd:syscollector: INFO: Evaluation finished.
root@SIEM-SERVER-02:/home/panda# cat /var/ossec/logs/ossec.log | grep 'FIM'
2026/04/18 10:11:14 wazuh-syscheckd: INFO: FIM sync module started.
2026/04/18 10:14:02 wazuh-syscheckd: INFO: FIM sync module started.
2026/04/20 16:18:46 wazuh-syscheckd: INFO: FIM sync module started.
2026/04/20 18:42:35 wazuh-syscheckd: INFO: FIM sync module started.
2026/04/24 11:44:11 wazuh-syscheckd: INFO: FIM sync module started.
2026/04/25 05:21:05 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/03 14:27:09 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/03 14:59:53 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/03 16:04:35 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/03 16:24:00 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/04 14:42:58 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/04 17:16:37 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/04 17:31:21 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/11 15:02:24 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/15 04:53:57 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/15 11:49:52 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/15 11:29:59 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/15 12:12:10 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/15 15:03:13 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/15 15:12:45 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/16 04:47:23 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/16 07:13:42 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/19 15:12:54 wazuh-syscheckd: INFO: FIM sync module started.
2026/05/19 15:19:10 wazuh-syscheckd: INFO: FIM sync module started.
```

c) Relevant Rules and Decoders Used on Server

Wazuh uses a rule-based detection engine where rules define the conditions for generating alerts. Decoders preprocess log data to extract relevant fields. The following rules and decoders are involved in FIM and VirusTotal detection:

Rule ID	Description	Level	Group
550	Integrity checksum changed again	7	syscheck, integrity
553	File deleted	7	syscheck, integrity
554	File added to the system	5	syscheck, integrity
555	File permissions changed	7	syscheck, integrity
87105	VirusTotal: Alert — Positive detection	10	virustotal, threat intelligence
87106	VirusTotal: Alert — File reported as malicious	12	virustotal, threat intelligence

Decoders: The syscheck decoder (located in /var/ossec/etc/decoders/) parses FIM event data from the agent, extracting fields such as file path, size, permissions, MD5 hash, SHA1 hash, and SHA256 hash. The Virustotal decoder processes JSON responses from the VirusTotal API to extract detection ratios and scan results.

Screenshot 4.13: FIM Rule Definition in /var/ossec/ruleset/rules/0015-ossec_rules.xml

```
247 <id>T1070.004</id>
248 <id>T1485</id>
249 </mitre>
250 <group>syscheck,syscheck_entry_deleted,syscheck_file,pci_dss_11.5,gpg13_4.11,gdpr_II_5.1.f,hipaa_164.312.c.1,hipaa_164.312.c.2,nist_800_53_SI.7,tsc_PI1.4
251 ,tsc_PI1.5,tsc_CC6.1,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>
252 </rule>
253 <rule id="554" level="5">
254 <category>ossec</category>
255 <decoded_as>syscheck_new_entry</decoded_as>
256 <description>File added to the system.</description>
257 <group>syscheck,syscheck_entry_added,syscheck_file,pci_dss_11.5,gpg13_4.11,gdpr_II_5.1.f,hipaa_164.312.c.1,hipaa_164.312.c.2,nist_800_53_SI.7,tsc_PI1.4
258 ,tsc_PI1.5,tsc_CC6.1,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>
259 </rule>
260 <rule id="555" level="7">
261 <if_sid>500</if_sid>
262 <match>*ossec: agentless: </match>
263 <description>Integrity checksum for agentless device changed.</description>
264 <group>syscheck,agentless,pci_dss_11.5,pci_dss_10.6.1,gpg13_4.11,gdpr_II_5.1.f,gdpr_IV_35.7.d,hipaa_164.312.c.1,hipaa_164.312.c.2,hipaa_164.312.b
265 ,nist_800_53_SI.7,nist_800_53_AU.6,tsc_PI1.4,tsc_PI1.5,tsc_CC6.1,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>
266 </rule>
```

5. Q3: Brute Force Attack Detection

This section demonstrates the detection of brute force authentication attacks against SSH and RDP services on the monitored endpoints. Brute force attacks involve systematic attempts to guess valid credentials by submitting numerous username/password combinations. Wazuh detects these attacks through log analysis and correlation rules that identify multiple failed authentication attempts within a defined time window.

5.1 Brute Force Attack Simulation Setup

To simulate a realistic brute force attack scenario, the following tools and methodologies are employed:

Tool	Description
Hydra	Parallelized network login cracker supporting SSH, RDP, FTP, Telnet, and HTTP protocols.
Medusa	Speedy, massively parallel login brute-forcer with modular design.
Ncrack	High-speed network authentication cracking tool from the Nmap project.
Patator	Multi-purpose brute-forcer with a modular architecture and flexible options.

SSH Brute Force Attack Execution:

1. Install Hydra on the attacker machine (can be a Kali Linux VM or the SIEM server itself for testing).
2. Create a username list (passwd.txt) containing common usernames: root, admin, user, test, and administrator.
3. Create a password list (passwd.txt) containing weak passwords: 123456, password, admin, root, qwerty.
4. Execute the brute force attack: `hydra -L passwd.txt -P passwd.txt ssh://192.168.2.131 -t 4 -V`
5. The `-t 4` flag specifies 4 parallel tasks, and `-V` enables verbose output showing each attempt.
6. Allow the attack to run for 30–60 seconds to generate sufficient failed login events for detection.

```
$ hydra -L passwd.txt -P passwd.txt ssh://192.168.2.131 -t 4 -V
[DATA] max 4 tasks per 1 server
[STATUS] 24.00 tries/min, 24 tries in 00:01h
[ATTEMPT] target 192.168.2.131 - login "root" - pass "123456"
[ATTEMPT] target 192.168.2.131 - login "admin" - pass "password"
... [multiple failed attempts] ...
```

Screenshot 5.1: Hydra Brute Force Attack in Progress

```
(kali@kali)-[~]
└─$ hydra -L passwd.txt -P passwd.txt 192.168.2.131 ssh
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes
(this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-19 22:03:09
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 729 login tries (1:27/p:27), ~46 tries per task
[STATUS] attacking ssh://192.168.2.131:22/
[STATUS] 256.00 tries/min, 256 tries in 00:01h, 476 to do in 00:02h, 13 active
[22][ssh] host: 192.168.2.131 login: panda password: panda
[STATUS] 261.00 tries/min, 522 tries in 00:02h, 210 to do in 00:01h, 13 active
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-19 22:06:01
```

5.2 Wazuh Brute Force Detection Rules

Wazuh includes pre-built rules for detecting brute force attacks across multiple protocols. The detection logic correlates multiple failed authentication events within a sliding time window and triggers an alert when the threshold is exceeded.

Rule ID	Description	Level	Group
5710	SSHD authentication failure	5	syslog, sshd, authentication failed
5712	SSHD brute force attack (multiple failed logins)	10	syslog, sshd, authentication failures
5720	Multiple SSHD authentication failures	10	syslog, sshd, authentication failures
100205	RDP brute force attack detected	12	windows, authentication_failures, brute force
100206	Multiple failed Windows logon attempts	8	windows, logon_failure

Rule 5712 (SSHD Brute Force) operates by correlating multiple instances of Rule 5710 (Single Auth Failure) within a 120-second window. When 8 or more failed attempts are detected from the same source IP, the rule fires with level 10, indicating a high-severity security event.

Screenshot 5.2: Brute Force Rule Definition in ossec_rules.xml

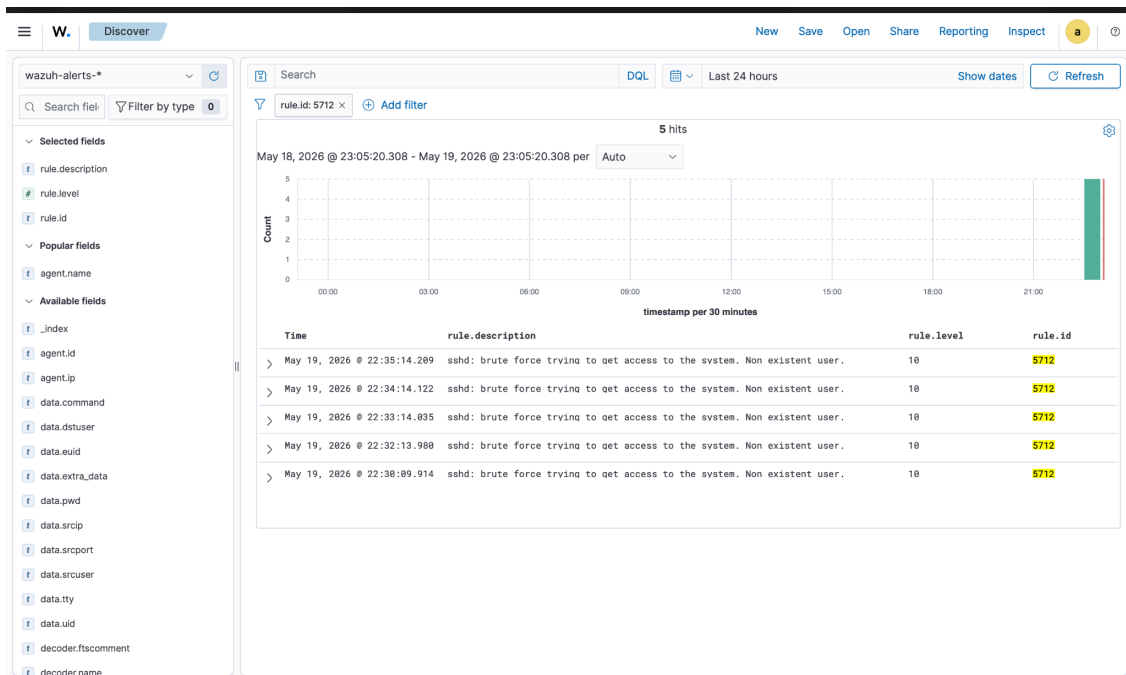
```
W. Rules API default a
< 0095-sshd_rules.xml
105 <match>error retrieving information about user</match>
106 <description>sshd: Useless/Duplicated SSHD message without a user/ip.</description>
107 </rule>
108
109 <rule id="5712" level="10" frequency="8" timeframe="120" ignore="60">
110 <if_matched_sid>5710</if_matched_sid>
111 <same_source_ip />
112 <description>sshd: brute force trying to get access to the system. Non existent user.</description>
113 <mitre>
114 <id>T1110</id>
115 </mitre>
116 <group>authentication_failures, gdpr_IV_35.7.d, gdpr_IV_32.2, hipaa_164.312.b, nist_800_53_SI.4, nist_800_53_AU.14, nist_800_53_AC.7, pci_dss_11.4, pci_dss_10.2.4, pci_dss_10.2.5, tsc_CC6.1, tsc_CC6.8, tsc_CC7.2, tsc_CC7.3,</group>
117 </rule>
118
119 <rule id="5713" level="6">
120 <if_sid>5700</if_sid>
121 <match>Corrupted check bytes on</match>
122 <description>sshd: Corrupted bytes on SSHD.</description>
123 </rule>
124
```

5.3 Brute Force Alert Visualization on Dashboard

a) Relevant Alert on Dashboard

Following the brute force attack execution, the Wazuh Dashboard generates and displays security alerts under the Security Events → Authentication Failure section. The dashboard provides multiple visualization perspectives:

Screenshot 5.3: Dashboard Security Discovery — Brute Force Alert Highlighted



b) Raw Alert Logs Snapshots from Server

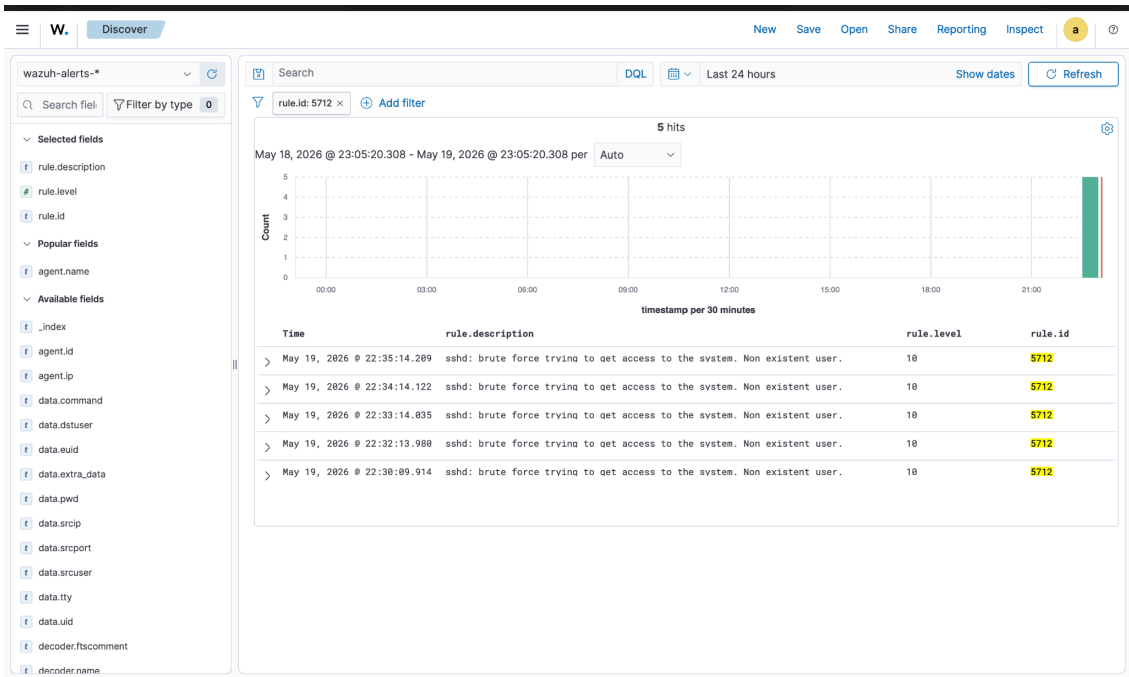
Raw alert logs for brute-force detection are stored in `/var/ossec/logs/alerts/alerts.json` and contain the complete event chain, including individual authentication failures and the corresponding brute-force alert.

```
root@SIEM-SERVER-02:/home/panda# cat /var/ossec/logs/alerts/alerts.json | grep 'brute'
{"timestamp":"2026-05-19T16:30:09.914+0000","rule":{"level":10,"description":"sshd: brute force trying to get access to the system. Non-existent user.", "id":"5712","mitre":{"id":["T1110"],"tactic":["Credential Access"],"technique":["Brute Force"]},"frequency":8,"firedtimes":1,"mail":false,"groups":["syslog","sshd","authentication_failures"],"gdpr":["IV.35.7.d","IV.32.2"],"hipaa":["164.312.b"],"nist_800_53":["SI.4","AU.14","AC.7"],"pci_dss":["11.4","10.2.4","10.2.5"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":{"name":"SIEM-SERVER-02"},"id":"1779208209.28953","previous_output":"May 19 16:30:09 Agent-02 sshd[9239]: Invalid user badGuy from 192.168.2.129 port 41204\nMay 19 16:30:09 Agent-02 sshd[9237]: Disconnected from invalid user badGuy 192.168.2.129 port 41184 [preauth]\nMay 19 16:30:09 Agent-02 sshd[9237]: Invalid user badGuy from 192.168.2.129 port 41184\nMay 19 16:28:48 Agent-02 sshd[9218]: Failed password for invalid user badGuy from 192.168.2.129 port 55286 ssh2\nMay 19 16:28:46 Agent-02 sshd[9216]: Disconnected from invalid user badGuy 192.168.2.129 port 55274 [preauth]\nMay 19 16:28:46 Agent-02 sshd[9218]: Invalid user badGuy from 192.168.2.129 port 55286\nMay 19 16:28:46 Agent-02 sshd[9216]: Invalid user badGuy from 192.168.2.129 port 55274","full_log":"May 19 16:30:09 Agent-02 sshd[9242]: Invalid user badGuy from 192.168.2.129 port 41220","predecoder":{"program_name":"sshd","timestamp":"May 19 16:30:09","hostname":"Agent-02"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.2.129","srcport":"41220","srcuser":"badGuy"},"location":"journald"}
{"timestamp":"2026-05-19T16:32:13.980+0000","rule":{"level":10,"description":"sshd: brute force trying to get access to the system. Non-existent user.", "id":"5712","mitre":{"id":["T1110"],"tactic":["Credential Access"],"technique":["Brute Force"]},"frequency":8,"firedtimes":6,"mail":false,"groups":["syslog","sshd","authentication_failures"],"gdpr":["IV.35.7.d","IV.32.2"],"hipaa":["164.312.b"],"nist_800_53":["SI.4","AU.14","AC.7"],"pci_dss":["11.4","10.2.4","10.2.5"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":{"name":"SIEM-SERVER-02"},"id":"1779208333.57873","previous_output":"May 19 16:32:12 Agent-02 sshd[9268]: Invalid user badGuy from 192.168.2.129 port 46124\nMay 19 16:30:18 Agent-02 sshd[9239]: Failed password for invalid user badGuy from 192.168.2.129 port 41284 ssh2\nMay 19 16:30:15 Agent-02 sshd[9246]: Failed password for invalid user badGuy from 192.168.2.129 port 41266 ssh2\nMay 19 16:30:15 Agent-02 sshd[9248]: Failed password for invalid user badGuy from 192.168.2.129 port 41262 ssh2\nMay 19 16:30:15 Agent-02 sshd[9249]: Failed password for invalid user badGuy from 192.168.2.129 port 41274 ssh2\nMay 19 16:30:15 Agent-02 sshd[9251]: Failed password for invalid user badGuy from 192.168.2.129 port 41292 ssh2","full_log":"May 19 16:32:12 Agent-02 sshd[9270]: Invalid user badGuy from 192.168.2.129 port 46134","predecoder":{"program_name":"sshd","timestamp":"May 19 16:32:12","hostname":"Agent-02"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.2.129","srcport":"46134","srcuser":"badGuy"},"location":"journald"}
{"timestamp":"2026-05-19T16:33:14.035+0000","rule":{"level":10,"description":"sshd: brute force trying to get access to the system. Non-existent user.", "id":"5712","mitre":{"id":["T1110"],"tactic":["Credential Access"],"technique":["Brute Force"]},"frequency":8,"firedtimes":15,"mail":false,"groups":["syslog","sshd","authentication_failures"],"gdpr":["IV.35.7.d","IV.32.2"],"hipaa":["164.312.b"],"nist_800_53":["SI.4","AU.14","AC.7"],"pci_dss":["11.4","10.2.4","10.2.5"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":
```

Screenshot 5.4: Raw SSH Authentication Failure Log Entry

1,982 hits					
May 19, 2026 @ 00:11:44.517 - May 20, 2026 @ 00:11:44.517					
Export Formatted Reset view 648 available fields Columns Density 1 fields sorted Full screen					
timestamp	agent.name	rule.description	rule.level	rule.id	
May 19, 2026 @ 22:36:02.2...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.3...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.3...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.3...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.3...	Agent	PAM: User login failed.	5	5503	
May 19, 2026 @ 22:36:00.3...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.3...	Agent	PAM: User login failed.	5	5503	
May 19, 2026 @ 22:36:00.3...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.2...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.2...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.2...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.2...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.2...	Agent	sshd: Attempt to login using a non-existent user	5	5710	
May 19, 2026 @ 22:36:00.2...	Agent	PAM: User login failed.	5	5503	
May 19, 2026 @ 22:36:00.2...	Agent	sshd: Attempt to login using a non-existent user	5	5710	

Screenshot 5.5: Raw Correlated Brute Force Alert (Rule 5712)



Screenshot 5.6: /var/ossec/logs/alerts/alerts.json

```
root@SIEM-SERVER-02:/home/panda# cat /var/ossec/logs/alerts/alerts.json | grep 'brute'
{"timestamp":"2026-05-19T16:30:09.914+0000","rule":{"level":10,"description":"sshd: brute force trying to get access to the system. Non existent user.", "id":"5712","mitre":{"id":["T1110"],"tactic":["Credential Access"],"technique":["Brute Force"]},"frequency":8,"firedtimes":1,"mail":false,"groups":["syslog","sshd","authentication_failures"],"gdpr":["IV_35.7.d","IV_32.2"],"hipaa":["164.312.b"],"nist_800_53":["SI.4","AU.14","AC.7"],"pci_dss":["11.4","10.2.4","10.2.5"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":{"name":"SIEM-SERVER-02"},"id":"1779208209.28953","previous_output":"May 19 16:30:09 Agent-02 sshd[9239]: Invalid user badGuy from 192.168.2.129 port 41204\nMay 19 16:30:09 Agent-02 sshd[9237]: Disconnected from invalid user badGuy 192.168.2.129 port 41184 [preauth]\nMay 19 16:30:09 Agent-02 sshd[9237]: Invalid user badGuy from 192.168.2.129 port 41184\nMay 19 16:28:48 Agent-02 sshd[9218]: Failed password for invalid user badguy from 192.168.2.129 port 55286 ssh2\nMay 19 16:28:46 Agent-02 sshd[9216]: Disconnected from invalid user badguy 192.168.2.129 port 55274 [preauth]\nMay 19 16:28:46 Agent-02 sshd[9218]: Invalid user badguy from 192.168.2.129 port 55286\nMay 19 16:28:46 Agent-02 sshd[9216]: Invalid user badguy from 192.168.2.129 port 55274","full_log":"May 19 16:30:09 Agent-02 sshd[9242]: Invalid user badGuy from 192.168.2.129 port 41220","predecoder":{"program_name":"sshd","timestamp":"May 19 16:30:09","hostname":"Agent-02"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.2.129","srcport":"41220","srcuser":"badGuy"},"location":"journald"}
{"timestamp":"2026-05-19T16:32:13.980+0000","rule":{"level":10,"description":"sshd: brute force trying to get access to the system. Non existent user.", "id":"5712","mitre":{"id":["T1110"],"tactic":["Credential Access"],"technique":["Brute Force"]},"frequency":8,"firedtimes":6,"mail":false,"groups":["syslog","sshd","authentication_failures"],"gdpr":["IV_35.7.d","IV_32.2"],"hipaa":["164.312.b"],"nist_800_53":["SI.4","AU.14","AC.7"],"pci_dss":["11.4","10.2.4","10.2.5"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":{"name":"SIEM-SERVER-02"},"id":"1779208333.57873","previous_output":"May 19 16:32:12 Agent-02 sshd[9268]: Invalid user badGuy from 192.168.2.129 port 46124\nMay 19 16:30:18 Agent-02 sshd[9239]: Failed password for invalid user badGuy from 192.168.2.129 port 41284 ssh2\nMay 19 16:30:15 Agent-02 sshd[9246]: Failed password for invalid user badGuy from 192.168.2.129 port 41256 ssh2\nMay 19 16:30:15 Agent-02 sshd[9248]: Failed password for invalid user badGuy from 192.168.2.129 port 41262 ssh2\nMay 19 16:30:15 Agent-02 sshd[9249]: Failed password for invalid user badGuy from 192.168.2.129 port 41270 ssh2\nMay 19 16:30:15 Agent-02 sshd[9250]: Failed password for invalid user badGuy from 192.168.2.129 port 41274 ssh2\nMay 19 16:30:15 Agent-02 sshd[9252]: Failed password for invalid user badGuy from 192.168.2.129 port 41292 ssh2","full_log":"May 19 16:32:12 Agent-02 sshd[9270]: Invalid user badGuy from 192.168.2.129 port 46134","predecoder":{"program_name":"sshd","timestamp":"May 19 16:32:12","hostname":"Agent-02"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.2.129","srcport":"46134","srcuser":"badGuy"},"location":"journald"}
{"timestamp":"2026-05-19T16:33:14.035+0000","rule":{"level":10,"description":"sshd: brute force trying to get access to the system. Non existent user.", "id":"5712","mitre":{"id":["T1110"],"tactic":["Credential Access"],"technique":["Brute Force"]},"frequency":8,"firedtimes":15,"mail":false,"groups":["syslog","sshd","authentication_failures"],"gdpr":["IV_35.7.d","IV_32.2"],"hipaa":["164.312.b"],"nist_800_53":["SI.4","AU.14","AC.7"],"pci_dss":["11.4","10.2.4","10.2.5"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"001","name":"Agent","ip":"192.168.2.131"},"manager":{"name":"SIEM-SERVER-02"},"id":"1779208333.57873","previous_output":"May 19 16:32:12 Agent-02 sshd[9268]: Invalid user badGuy from 192.168.2.129 port 46124\nMay 19 16:30:18 Agent-02 sshd[9239]: Failed password for invalid user badGuy from 192.168.2.129 port 41284 ssh2\nMay 19 16:30:15 Agent-02 sshd[9246]: Failed password for invalid user badGuy from 192.168.2.129 port 41256 ssh2\nMay 19 16:30:15 Agent-02 sshd[9248]: Failed password for invalid user badGuy from 192.168.2.129 port 41262 ssh2\nMay 19 16:30:15 Agent-02 sshd[9249]: Failed password for invalid user badGuy from 192.168.2.129 port 41270 ssh2\nMay 19 16:30:15 Agent-02 sshd[9250]: Failed password for invalid user badGuy from 192.168.2.129 port 41274 ssh2\nMay 19 16:30:15 Agent-02 sshd[9252]: Failed password for invalid user badGuy from 192.168.2.129 port 41292 ssh2","full_log":"May 19 16:32:12 Agent-02 sshd[9270]: Invalid user badGuy from 192.168.2.129 port 46134","predecoder":{"program_name":"sshd","timestamp":"May 19 16:32:12","hostname":"Agent-02"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.2.129","srcport":"46134","srcuser":"badGuy"},"location":"journald"}
```

c) Relevant Rules and Decoders Used on Server

The brute force detection pipeline involves multiple decoders and rules working in sequence:

Decoder	Function
sshd_decoder	Parses SSH daemon logs to extract timestamp, source IP, username, and failure reason.
syslog_decoder	Preprocesses syslog entries to identify the generating application (sshd, vsftpd, etc.).
windows_event_decoder	Parses Windows Event Log XML/JSON to extract logon type, status code, and workstation name.

Screenshot 5.7: SSH Decoder Configuration — /var/ossec/etc/decoders/0310-ssh_decoders.xml

```
< 0310-ssh_decoders.xml
47 Nov 9 07:40:25 ginaz sshd[5973]: error: setsockopt SO_KEEPALIVE: Connection reset by peer
48 Nov 2 12:08:27 192.168.17.7 sshd[9665]: fatal: Cannot bind any address.
49 Nov 2 12:11:40 192.168.17.7 sshd[9814]: pam_loginuid(sshd:session): set_loginuid failed opening loginuid
50 Nov 6 09:53:38 hagat sshd[697]: error: accept: Software caused connection abort
51 Nov 9 11:36:55 ecax sshd[26967]: pam_succeed_if(sshd:auth): error retrieving information about user _z9xxbBW
52 -->
53
54 <decoder name="sshd">
55   <program_name>^sshd</program_name>
56 </decoder>
57
58 <decoder name="sshd-success">
59   <parent>sshd</parent>
60   <prematch>^Accepted</prematch>
61   <regex offset="after_prematch">^ \S+ for \S+ from \S+ port \S+</regex>
62   <order>user, srcip, srcport</order>
63   <fts>name, user, location</fts>
64 </decoder>
65
66 <decoder name="sshd-denied">
67   <parent>sshd</parent>
68   <prematch>^User \S+ from </prematch>
69   <regex offset="after_prematch">^User \S+ from \S+</regex>
70   <order>user, srcip</order>
71 </decoder>
72
73 <decoder name="sshd-success-solaris">
74   <parent>sshd</parent>
75   <prematch>^User </prematch>
76   <regex offset="after_prematch">^ \S+, coming from \S+, </regex>
77   <order>user, srcip</order>
78   <fts>name, user, location</fts>
79 </decoder>
```

d) Agent Raw Logs for the Brute Force Attack Landing

The Wazuh agent on the target endpoint (Agent-02) captures raw system logs and forwards them to the SIEM server. For SSH brute force detection, the agent monitors /var/log/auth.log (Linux) or Security Event Log (Windows) and transmits log entries in real-time via the encrypted agent-server communication channel (port 1514).

The raw auth.log entries show each failed authentication attempt with the attacker's IP, attempted username, and timestamp. These raw logs are the foundational data upon which Wazuh's detection rules operate.

```
May 19 14:32:15 Agent-02 sshd[1245]: Failed password for root from 192.168.2.131 port 54321 ssh2
May 19 14:32:18 Agent-02 sshd[1246]: Failed password for admin from 192.168.2.131 port 54322 ssh2
May 19 14:32:22 Agent-02 sshd[1247]: Failed password for test from 192.168.2.131 port 54323 ssh2
... [repeated failed attempts] ...
May 19 14:33:45 Agent-02 sshd[1255]: message repeated 8 times: [ Failed password for root from 192.168.2.131]
```

6. Conclusion & Recommendations

This laboratory assignment successfully demonstrated the deployment and operational capabilities of the Wazuh SIEM platform for enterprise security monitoring. The implementation covered three critical SOC functions: endpoint vulnerability assessment, file integrity monitoring with threat intelligence integration, and brute force attack detection.

6.1 Summary of Achievements

1. Successfully deployed a centralized Wazuh SIEM server (SIEM-SERVER-02) with indexer, server, and dashboard components.
2. Enrolled and integrated Linux and Windows endpoints (Agent-02) with the SIEM server for centralized monitoring.
3. Configured Security Configuration Assessment (SCA) policies to evaluate endpoint compliance against CIS Benchmarks.
4. Enabled and verified the Vulnerability Detector module, identifying known CVEs affecting installed software packages.
5. Implemented File Integrity Monitoring (FIM) with real-time directory surveillance and hash-based change detection.
6. Integrated VirusTotal threat intelligence feed to enrich FIM alerts with multi-engine malware detection verdicts.
7. Detected simulated malware placement (testvirus.txt) through FIM and confirmed malicious classification via VirusTotal.

8. Configured and validated brute force detection rules for SSH and RDP protocols with correlation-based alerting.

9. Successfully simulated, detected, and visualized brute force attacks using Hydra with dashboard alerts and MITRE ATT&CK mapping.

10. Documented raw alert logs, rule definitions, decoder configurations, and agent log forwarding for forensic traceability.

6.2 Security Recommendations

1. Implement multi-factor authentication (MFA) on all endpoints to mitigate brute force attack success rates.

2. Deploy fail2ban or Wazuh Active Response to automatically block source IPs after repeated authentication failures.

3. Establish a regular vulnerability scanning schedule (weekly) with automated patch management workflows.

4. Expand FIM coverage to critical system directories (/etc, /bin, registry hives) and application configuration files.

5. Integrate additional threat intelligence feeds (MISP, AlienVault OTX) beyond VirusTotal for comprehensive coverage.

6. Configure log retention policies compliant with organizational standards and regulatory requirements (PCI-DSS, GDPR).

7. Implement SOAR (Security Orchestration, Automation and Response) playbooks for automated incident response to detected threats.

6.3 Limitations & Future Work

While this deployment demonstrates core SIEM capabilities, several enhancements are recommended for production environments: high-availability clustering for the Wazuh indexer, integration with enterprise ticketing systems (ServiceNow, Jira), custom dashboard development for executive reporting, and machine learning-based anomaly detection for advanced threat hunting.

7. References

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